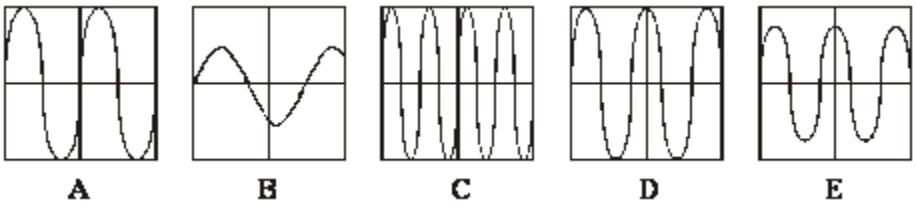
Q1.

(a) A student uses a microphone to send different sounds to an oscilloscope. The diagrams show five traces, **A**, **B**, **C**, **D** and **E**, on the oscilloscope. All the traces are drawn to the same scale.



(i) Which **three** diagrams show traces with the same amplitude?

Diagrams ..... , ... and ..... .

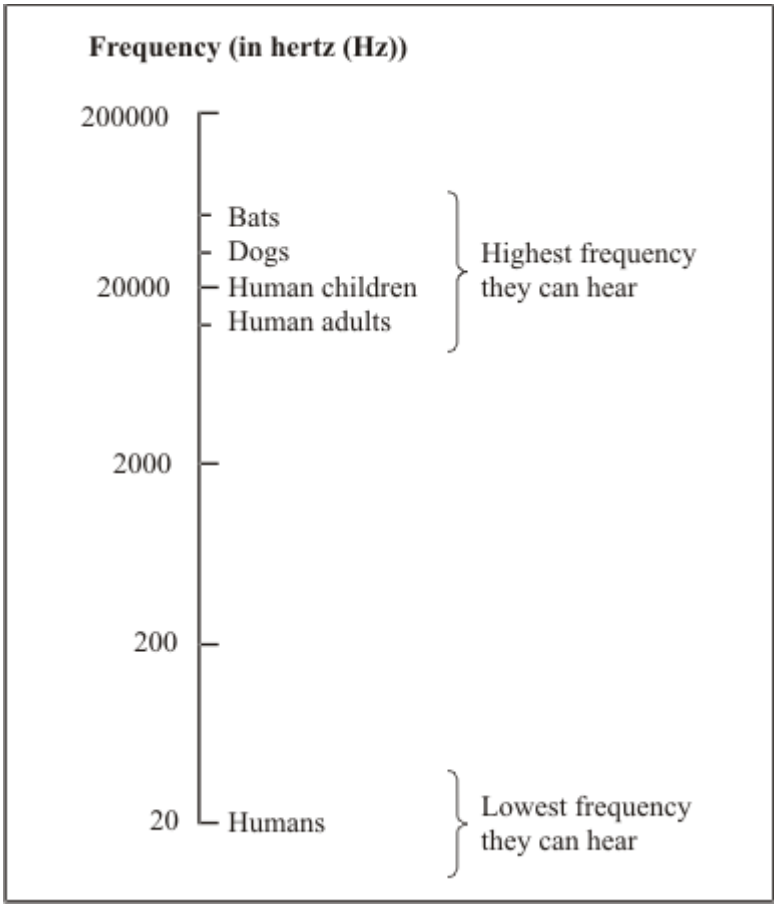
(1)

(ii) Which **two** diagrams show traces with the same frequency?

Diagrams ..... and ..... .

(1)

(b) The diagram shows the sound frequencies which some living things can hear.



(i) What is the widest range of frequencies that a human child can hear?

_____

(1)

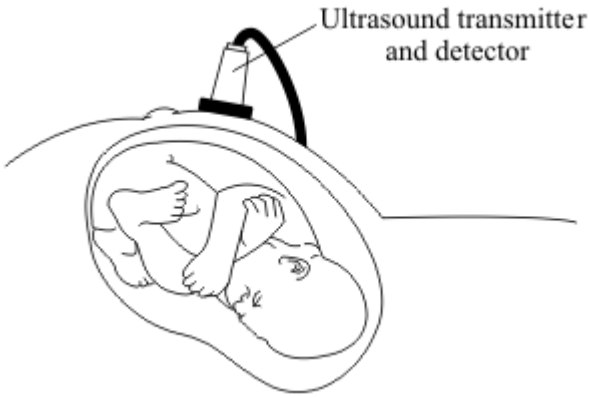
(ii) Why can some dog whistles be heard by dogs but not by humans?

_____

_____

(1)

(c) An ultrasound scan can be used to make a picture of a baby in its mother's womb. An ultrasound transmitter and detector are placed above the mother's womb. Ultrasound goes into the body of the mother and into the body of the baby.



Use the correct words from the box to complete the sentences.

detector	**reflection**	**refraction**	**sound**	**substance**	**transmitter**

(i) When the ultrasound crosses from one ..... to another, some ultrasound becomes an echo caused by ..... .

(ii) This information is collected by the ultrasound ..... and made into a picture on a screen.

(3)

(Total 7 marks)

Q1.

(a) (i) **A**, **C** and **D**

any order but all three required and no others

1

(ii) **D** and **E**

either order but both required and no others

1

(b) (i) 20000 (Hz) to 20 (Hz)

accept '19980 (Hz)'

or vice-versa

1

(ii) frequency (of dog whistle) too high (for humans to hear) / frequency above 20000 Hz

accept 'it is ultrasound'

accept 'sound from the whistle is ultrasonic'

1

(c) (i) substance

1

reflection

correct order essential

1

(ii) detector

1

[7]